

Canonical notation, identifiers, and glossary

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This file is the single source of truth for mathematical notation, code identifiers, and controlled vocabulary across the manuscripts under `analysis/report/`. New manuscripts should adopt it rather than re-deriving symbols, and any deviation should be stated explicitly in the deviating paper's Methods section.

The audit behind it is `analysis/report/whitepaper-notation-audit.md`, whose PDF binds this file in as its appendix so that the two cannot drift apart.

Symbols and identifiers listed here were extracted from the manuscript sources, from R/, from the drivers under `analysis/scripts/`, and from the archived `.rds` summaries.

Part 1. Mathematical symbols

Indices and structure

Symbol	Meaning
i	participant index
t	timepoint index; weekly measurement occasions throughout
j, ij	period or occasion index within participant (papers 04, 06)
N	total number of participants in the trial, across all randomization paths; never a per-path count. See Part 2, sample size and randomization paths
P	number of randomization paths in a design; the expected allocation to each is N/P
$n_{\text{reps}}, n_{\text{sim}}$	Monte Carlo replicates per cell
k	cycles per participant in design sweeps; also the Weibull shape parameter in the decay family

Outcome and latent components

Symbol	Meaning
Y_{it}	observed symptom score; the modelled outcome (code column <code>Sx</code>)
BL_i	participant-specific baseline level
BR_{it}	biological (pharmacological) response component
PB_{it}	placebo-belief response component
TV_{it}	time-variant natural-history component
ε_{it}	observation-level residual
u_i	participant random intercept
η	expectancy covariate (measured belief), paper 06

Treatment exposure and carryover

Symbol	Meaning
D_{it}	binary drug state: 1 on drug, 0 off drug
$D_{bc,it}$	continuous exposure-decayed drug indicator: 1 on drug, $e^{-\lambda t_{sd}}$ off drug (code column <code>Dbc</code>)

Symbol	Meaning
t_{sd}	time since discontinuation; input to the carryover decay (code <code>tsd</code>)
t_{od}	cumulative time on drug; input to the Gompertz response (code <code>tod</code>)
t_{pb}	time-by-placebo-belief accumulation (code <code>tpb</code>)
$t_{1/2}$	carryover half-life (code <code>carryover_t1half</code>)
λ	carryover decay rate, $\ln 2/t_{1/2}$ (code <code>lambda_cor</code>)
λ_w	Weibull rate, $(\ln 2)^{1/k}/t_{1/2}$

Decay families implemented on the data-generating side: exponential $e^{-\lambda t}$; linear $\max(0, 1 - t/(2t_{1/2}))$; Weibull $e^{-(\lambda_w t)^k}$; power $\max(0, (1 - t/(3t_{1/2}))^p)$.

Biomarker and moderation

Symbol	Meaning
B_i	participant-level pre-treatment biomarker value (code <code>bm</code>)
b_i	standardized biomarker, $(B_i - \bar{B})/s_B$ (code <code>bm_z</code>)
c_{bm}	moderation parameter of the covariance-moderation architecture: on-drug correlation between biomarker and BR . A correlation, bounded in $[-1, 1]$
β_{bm}	moderation parameter of the mean-moderation architecture: dimensionless multiplier scaling the treatment effect by b_i
$c_{bm,A}, c_{bm,B}$	independent mean-channel and covariance-channel weights of the dual-channel architecture
σ_{BR}, σ_{bm}	standard deviations of the response component and of the biomarker

Estimand, inference, and performance

Symbol	Meaning
$\beta_{bm:D}$	the target estimand: the biomarker-by-treatment interaction coefficient (<code>bm:Dbc</code>)
$\beta_{bm}^{BR}, \beta_{bm}^{PB}, \beta_{bm}^{TV}$	component-specific biomarker slopes (paper 06)
$\beta_{bm}^{\text{lumped}}$	the one-component (single-indicator) biomarker slope
β_D	treatment main effect (papers 04, 05)
$\theta, \hat{\theta}$	a generic interaction coefficient; used only where the argument is deliberately not N-of-1 specific (paper 10)
θ_{true}	calibrated true value, $-c_{bm} \sigma_{BR}$
ρ	AR(1) within-factor serial correlation
κ	ratio of mean model-based standard error to empirical standard deviation; $\kappa > 1$ means a conservative test
δ	standardized bias of the test statistic
ν	denominator degrees of freedom
α	nominal significance level; 0.05 throughout
π	power, $\Pr(p < \alpha)$
γ	latent-class gating slope (paper 03); a $BR \times PB$ interaction coefficient (paper 06)

Symbol	Meaning
Z_i	unobserved latent class label (paper 03)

The five rules the symbols encode

1. D_{it} **is binary**; $D_{bc,it}$ **is continuous**. These are different objects, and the distinction is the subject of the carryover-specification manuscript. Never use one for the other.
2. c_{bm} **is a correlation**; β_{bm} **is a multiplier**. They are calibrated to a common numeric scale, because the mean-channel shift is applied as $\beta_{bm} b_i \sigma_{BR}$ on the standardized biomarker, so the reference value 0.45 denotes a matched moderation strength under either architecture. The symbols are nonetheless not interchangeable. A paper reporting both architectures at a matched value may use one shared label, but must say so.
3. **Mathematics in model definitions, code identifiers in code contexts**. Write $Y_{ij} = \beta_0 + \beta_D D_{bc,ij} + \dots$ for the model; write `Sx ~ bm + t + Dbc + bm:Dbc` in `\texttt{}` when quoting the R formula. Never mix the two inside one expression.
4. **N is the total sample size**. N is the total number of participants across all randomization paths, and the expected allocation to each of P paths is N/P : with $P = 4$ and $N = 140$, each path receives about 35. Never use N for a per-path count. A paper may name a per-path count alongside the total, in words or as N/P , but must give N itself as the total. Because two of the drivers fix the per-path count instead, the reported N cannot be taken on trust: check the driver against the table in Part 2 before quoting a sample size.
5. **State the sign convention**. The three components are non-negative *reductions* in symptom severity, so an increase in any component lowers Y . Treatment effects and interaction coefficients are consequently negative; moderation parameters are positive.

What the linter checks

`tools/notation-lint.pl` mechanizes the part of this file that can be checked from the manuscript sources. Run it over the manuscripts:

```
perl tools/notation-lint.pl analysis/report/*/report.Rmd
```

It skips fenced R chunks and `verbatim` blocks and strips `\texttt{}` and backtick spans, so it sees reader-facing prose and mathematics only. Four checks are implemented.

Check	Fires on	Rule
<code>bare-Dbc</code>	<code>Dbc</code> as a symbol outside a code span	1
<code>bare-Sx</code>	<code>Sx</code> as a symbol outside a code span	3
<code>legacy-spec-label</code>	<code>A1</code> , <code>A2</code> , <code>A3</code>	Part 3, specifications
<code>N-per-path</code>	a value of N directly qualified as per-path	4

`N-per-path` fires on `$N = 35$ per randomization path` but not on `$N = 140$ (70 per path)`, which is the compliant way to give both numbers. A day-denominated $t_{1/2}$ is exempted when the file carries a `**Units.**` paragraph, which is how rule 3's units requirement is enforced.

Rule 2 and rule 5 are not mechanized: whether a shared c_{bm} label is declared, and whether the sign convention is stated, both need a reader. One known false positive stands, a `bare-Sx` inside a backtick span that crosses a line break in paper 02.

Units

Carryover half-life $t_{1/2}$ is quoted in **weeks** in the interaction-focused manuscripts, on the canonical grid $\{0, 0.5, 1.0\}$, and in **days** in the main-effect and test-procedure manuscripts, on a pharmacokinetic scale with a 3-day baseline. Both conventions are retained because each paper's figures are drawn on its own scale. Every paper using days carries an explicit units paragraph giving the conversion (1 week = 7 days). Timepoints t are weekly in both conventions.

Part 2. Code identifiers and stored data values

These are the names as they appear in R/, in the drivers, and in the archived .rds summaries.

Where a stored value is also a reporting label, the two are kept **identical**, so that no display-time mapping stands between the data and the page. A mapping layer is a standing hazard: it has to be applied at every display site, it fails silently when a rename reaches one layer and not the other, and it makes a stored table unreadable without the manuscript beside it. Where a stored value is a package API argument (`dgp_architecture = 'mvn'`), it stays as the API defines it and the manuscript maps it to a reporting name, which is the ordinary factor-level-to-label relationship.

Exported package functions

`buildtrialdesign`, `buildSigma`, `generateData`, `censordata`, `lme_analysis`, `generateSimulatedResults`, `validateParameterGrid`, `cumulative`, `modgompertz`, `trajectoryShape`, `shape_logistic`, `shape_hyperbolic_tangent`, `shape_piecewise_linear_breakpoint`, `characterize_carryover`, `analyze_trial_extended`, `print_carryover_summary`, `print_trial_summary`, `plotfactortrajectories`, `PlotModelingResults`, `reknitsimresults`.

Principal arguments

Identifier	Role
<code>modelparam</code>	list: sample size <code>N</code> , moderation <code>c.bm</code> . The meaning of <code>N</code> depends on the caller; see sample size and randomization paths below
<code>respparam</code> <code>blparam</code>	Gompertz parameters <code>maxr</code> , <code>rate</code> , <code>disp</code> per component biomarker and nuisance means and standard deviations
<code>trialdesign</code>	output of <code>buildtrialdesign</code> : <code>timepoints</code> , <code>timeptnames</code> , <code>expectancies</code> , <code>ondrug</code>
<code>dgp_architecture</code> <code>carryover_t1half</code>	'mvn', 'mean_moderation', or 'combined' carryover half-life
<code>lambda_cor</code> <code>br_family</code> <code>br_p2</code> , <code>br_p3</code>	decay rate; derived from the half-life when NA response-curve family: 'gompertz' and alternatives shape parameters of the alternative families
<code>moderation_scaling</code> <code>cached_sigma</code>	'constant' or trajectory-scaled pre-built covariance matrix, reused across parallel workers
<code>makePositiveDefinite</code> <code>n_cores</code> <code>save_chunks</code>	positive-definiteness repair switch worker count; -1 auto-detects progressive checkpointing for long runs

Sample size and randomization paths

Rule 4 fixes N as the total. The complication is that `modelparam$N` does not mean the same thing at every level of the call stack, and two families of driver read it differently.

The two levels. `generateData` generates one path: it takes a single `trialdesign` and treats N as the count for that path alone. `generateSimulatedResults` generates a whole design: it takes N as the total and allocates it, `Ns <- N %/% nP` with the remainder spread one per path. A driver that loops over `td$trialpaths` itself is therefore working at the lower level and must divide the total before the loop. A driver that passes the same full N to every iteration generates $N \times P$ participants, and its reported N is a per-path count wearing the wrong name.

Path counts. P is a property of the design, not a free parameter.

Design	P	Paths
OL	1	on drug throughout
CO	2	drug-then-placebo, placebo-then-drug
OL+BDC	2	discontinuation at either of two visits
Hybrid	4	two discontinuation points crossed with two crossover orders

Paper 06 also uses a reduced single-path OL+BDC variant for its smaller cells and a two-path sixteen-visit variant for its larger ones, so P changes within that paper’s sample-size axis.

Which drivers allocate and which duplicate. Established by inspection of every driver reachable from a current manuscript, on 2026-08-05.

Driver	Reads N as	Mechanism
R/generateSimulatedResults.R	total	<code>Ns <- N %/% nP</code> plus remainder
analysis/scripts/carryover-sensitivity/simulation-core.R	total	<code>generate_data_multi_path</code> , <code>floor(N / n_paths)</code>
implementations/noflpower/R/simulation.R	total	same integer split
analysis/scripts/test-procedure-design-sensitivity/01-study1-test-procedure.R	total	explicit <code>split_N</code>
analysis/scripts/quick-sim/*.R	per path	hand-rolled for <code>(g in seq_along(paths))</code> passing one mp
analysis/scripts/component-decomposition/*.R	per path	same shape
analysis/scripts/gompertz-evaluation/02-faithful-trajectory-sweep.R	per path	same shape

Status by paper. The manuscripts now report totals throughout. The column below records how each paper’s own driver behaves, because that is what a re-run will inherit. One entry is weaker than the others: the driver behind paper 01’s Section 3.3 robustness run was not located, so its per-path reading is taken from that paper’s own statement of its convention rather than from inspected code.

Paper	Driver reads N as	Manuscript
01	per path	relabeled to totals: 70 for CO and OL+BDC, 140 for Hybrid; the Section 3.3 check is OL+BDC at 140
02, 05, 08, 10, 11	total	already correct
03, 04	coincident (single-path or path-free DGP)	unaffected
06	per path	larger cells relabeled 100 to 200 and 150 to 300
07	per path	relabeled 35 to 70
09	per path	totals stated per design: 35, 70, 70, 140

Two consequences that outlive the relabeling. A per-path driver run across designs with different P produces an unmatched comparison, since the designs then differ in sample size as well as structure. This affects

the cross-design readings in papers 01 and 09, both of which now say so in place; within-design contrasts, which hold N fixed, are unaffected. Separately, the stored **N** column in the `.rds` output of the per-path drivers still holds per-path values, so a manuscript total and its stored **N** disagree for papers 01, 06, 07 and 09 until those drivers are corrected and the cells re-run. No current manuscript reads that column, so nothing renders wrong today.

Data columns

ptID (participant), **Sx** (outcome, the Y_{it} above), **bm** (biomarker), **bm_z** (standardized biomarker), **D_{bc}** (continuous exposure indicator, $D_{bc,it}$), **tsd**, **tod**, **tpb**, **t_{wk}** (time in weeks), **path** (randomization path).

Stored factor levels and their reporting labels

Column	Stored values	Reported as	Mapped?
spec	E1, E2, E3	E1, E2, E3	no, identical
carryover_form	exponential, linear, weibull	as stored	no, identical
design	CO, Hybrid, 0LBDC	CO, Hybrid, OL+BDC	only 0LBDC to OL+BDC
dgp_arch	mean_moderation, mvn, combined	Mean, Covariance, Dual	yes; these are package API values

The **spec** column stored A1, A2, A3 until 2026-07-31. Twenty-one `.rds` files were migrated to E1/E2/E3 and the producing drivers updated to emit them, so the display-time mapping that previously bridged the two has been retired. Pre-migration copies are in `analysis/.spec-migration-backup/`. Output archived before that date, including anything under a paper’s own `archive/`, still uses the old values.

Summary-table columns

n_reps, **n_converged**, **converged_frac**, **non_convergence_rate**, **power**, **bias**, **mean_estimate**, **true_value**, **empirical_se**, **mse**, **coverage**, and an **mc_se**-prefixed Monte Carlo standard error for each (**mc_se_power**, **mc_se_bias**, **mc_se_mse**, **mc_se_coverage**, **mc_se_non_convergence**).

Part 3. Controlled vocabulary and glossary

The project lexicon at `docs/29-nof1-precision-medicine-lexicon.md` holds the full terminology set (241 terms across 13 thematic sections, with cross-field synonyms). The entries below are the subset carrying a fixed, enforced meaning in this compendium.

Trial designs

OL (open-label). Unblinded titration; no off-drug contrast, so it supplies no within-subject interaction information. **BDC** (blinded discontinuation). A blinded switch to placebo after open-label response. **OL+BDC**. Open-label titration followed by blinded discontinuation. **CO** (crossover). Traditional two-period within-subject crossover. **Hybrid**. The Hendrickson (2020) design: open-label titration, blinded discontinuation, then a brief crossover. The programme’s reference design. Capitalize it. **Aggregated N-of-1 trial**. Many single-patient on/off series pooled into one mixed model.

Define each abbreviation at first use in every paper. Each design’s randomization-path count is given in Part 2, under sample size and randomization paths, and is not repeated here.

Data-generating architectures

Named by the channel that carries the interaction. Single letters are no longer used.

Name in prose	Legends and tables	Data value	Former label
the mean-moderation architecture	Mean	<code>mean_moderation</code>	Architecture A
the covariance-moderation architecture	Covariance	<code>mvn</code>	Architecture B
the dual-channel architecture	Dual	<code>combined</code>	Architecture C

Mean-moderation. The biomarker scales the treatment effect in the conditional mean; the interaction lives in the first moment. The near-universal convention in the trial-simulation literature. **Covariance-moderation.** The biomarker-response correlation is treatment-state dependent; the interaction lives in the second moment. The Hendrickson formulation, and the one whose power is most eroded by carryover. **Dual-channel.** Both channels active with independent weights $c_{bm,A}$ and $c_{bm,B}$; the other two are its single-channel limits. A data-generating construct only: the two weights are not separately identifiable from a fitted `bm:Dbc` coefficient.

Analysis specifications for carryover

E1. Binary on-drug indicator; carryover ignored. **E2.** Exposure-weighted continuous predictor D_{bc} , committing to an assumed decay form and half-life. **E3.** Binary indicator plus a lagged just-off-drug nuisance term; the classical Jones-Kenward crossover device, assuming no decay form.

The letter E stands for the exposure regressor that distinguishes them. These were labeled A1/A2/A3 until 2026-07-31; the stored data and the drivers were migrated to E1/E2/E3, so the labels now agree everywhere and nothing is mapped at display time.

Why not A/B/C and M1/M2/M3. The original scheme used A/B/C for architectures and A1/A2/A3 for specifications, colliding on the letter A within a single manuscript. Relabeling the specifications M1/M2/M3 removed that collision but created a second one: the calibration manuscript already uses M0-M3 for its working-covariance model ladder, and those labels are keys into archived data. E1/E2/E3 is free across the whole corpus, and naming the architectures by mechanism removes the class of collision entirely.

Response-curve families

Modified Gompertz (default), $y = \text{maxr} \cdot \exp(-\text{disp} \cdot \exp(-\text{rate} \cdot t))$ with a vertical offset so the curve passes through the origin. Alternatives: **logistic** (symmetric sigmoidal), **hyperbolic tangent** (fast-saturating), **piecewise-linear breakpoint** (no smoothness assumption).

Missing data and dropout

MCAR, MAR, MNAR in the standard Rubin sense. **Informative dropout.** Hazard depending on cumulative symptom worsening since baseline. **Happy accident.** The randomization-path selection effect by which informative dropout preferentially preserves the most informative crossover blocks.

Inference

LME. Linear mixed-effects model; the programme’s default analysis. **corCAR1.** Continuous-time AR(1) residual correlation structure. **GEE.** Generalized estimating equations; the marginal comparator. **Mancl-DeRouen, CR2.** Small-sample bias-corrected sandwich variance estimators. **RM-ANOVA.** Repeated-measures analysis of variance; the strict form requires a dichotomized biomarker. **Anti-conservative.** Realized Type I error above nominal. **Working covariance.** The assumed residual covariance of the LME; reserve **working correlation structure** for the GEE object.

Estimands and reporting

Biomarker-treatment interaction. The programme’s core estimand; equivalently the patient-by-treatment interaction, or heterogeneity of treatment effects (HTE) in the precision-medicine literature. **Predictive biomarker.** Predicts differential benefit, as opposed to a **prognostic** biomarker, which predicts outcome regardless of treatment. **ADEMP.** The Morris, White and Crowther (2019) reporting framework: aims, data-generating mechanisms, estimands, methods, performance measures. **MCSE.** Monte Carlo standard error; reported with every performance measure.

Enumeration conventions

Simulation studies. Number them (Study 1, Study 2, ...) rather than lettering them. **Sensitivity sweeps.** Labels S1, S2, ... are paper-local and are not comparable across papers. Cite them with the paper prefix when crossing manuscripts (05-S2, not S2).

Prose terminology

Symbol conventions are settled here; prose terminology is settled by `docs/30-terminology-consistency-plan.md`, executed 2026-07-30 with its D1 spelling decision locked to **US English**, overriding that document’s original frequency-based recommendation of British forms. The two are complementary: this file governs mathematics and identifiers, that one governs surface forms and acronym expansion.

Rendered on 2026-08-06 at 11:58 PDT. Source: ~/prj/res/36-pmsimstats-ng/pmsimstats-ng/analysis/report/NOTATION.md